

Altium Designer

This month's DVD includes Altium Designer—an EDA software package for printed circuit board, FPGA and embedded software design. It is developed and marketed by Altium Limited of Australia. Here we take a look at its features

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It is often felt that as manufacturing processes migrate to the region with the lowest cost, developed economies suffer. However, another way to look at it is that by outsourcing lower-value tasks, the resources previously occupied with them are now presented with much greater opportunities.

Design draws the same analogy: By employing design automation tools, engineers are able to increase their productivity. Today, engineers don't even think of a non-computerised approach to designing a PCB.

Frank Kramer, technical marketing director, EMEA, Altium Europe GmbH, says, "For electronic design engineers, use of EDA tools has led to vast improvements in the design process, from the component all the way down to the end product. Along with accelerating the design process, it is now possible to simulate or emulate every part of an electronic design before a single component is even purchased. In the field of IC development, for example, the very last step in a long and expensive process is to commit—or 'tape out'—a design to silicon. This, however, is largely due to the cost involved in manufacturing integrated devices—a cost that is only borne by the opportunity of selling the final device in large volumes."

Altium Designer presents hardware engineers, experienced FPGA engineers and embedded software developers a genuine opportunity to perform embedded system design

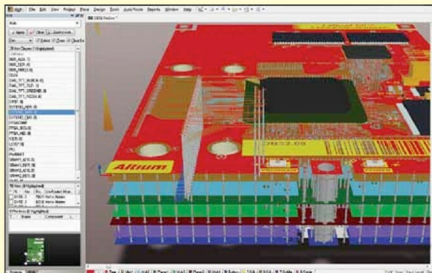


Fig. 1: Designing modern electronic products demands greater visibility and manipulation of multiple components

in an FPGA, moving it to a state of achievable reality from an inaccessible dream.

Key features of Altium Designer

Advances in EDA tools have made it easier to simulate a product's form fit and function before the product is manufactured. Regardless of this fact, with the obvious exception of IC design, the progress of design automation tools in the electronics sector has maintained focus on niche applications and vertical markets.

PCB design. Altium Designer provides a single application environment for capturing design data, verifying the performance of the circuit and designing the PCB, thereby unifying the process of board-level design. The single system and single model of

the design data redefine the way designers work by enabling single-point data management, real-time design synchronisation and smooth design collaboration with all domains.

Signal integrity. Altium Designer puts signal integrity directly into the PCB design environment. It is completely integrated with the board-level design rules to alert you of signal integrity issues as you design. You can run an analysis directly from the schematic or PCB design editors at any time. Altium Designer's signal integrity analyser uses transmission line calculations and input/output (I/O) buffer macro-model information as input for simulations.

FPGA design. Today, devices have sufficient capacity to implement more than just some of the hardware in a product—they can be programmed